Conversion of Camphene to Camphor Day 3

Grant Weatherford

Partner: None

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Introduction

In this experiment, borneol is combined with various chemicals, including glacial acetic acid, acetone, aqueous sodium hypochloride, methylene chloride, and saturated sodium bicarbonate to produce the final product, camphor. Slow and controlled additions of chemicals this week are important due to most of the reactions being exothermic. A powdery white camphor will be obtained at the end of the experiment that can be used to gather a percentage of yield calculation.

Materials and Methods

To start day three of the experiment, in a 10 mL long-neck round-bottom flask, 0.3 g of borneol product was added with 0.3 mL of glacial acetic acid. The racemic mixture was then dissolved in 1 mL of acetone and a boiling chip was added. Just like in day three of the experiment a reflux was performed with the distillation column and water bath. The water bath was kept between 40 and 50 degrees Celsius. Once all the reagents were dissolved, 6 mL of aqueous sodium hypochloride solution (bleach) was added to the flask through the top of the condenser dropwise for the next 35 minutes. After each addition of bleach, the mixture was gently swirled to ensure a thorough mix. The reaction was then left alone for 15 minutes after the last addition of bleach. The mixture was then taken out of the water bath and allowed to cool to room temperature. Once the mixture had reached room temperature it was transferred to a centrifuge tube. The camphor was then extracted with 4 mL of methylene chloride in two portions. The combined methylene chloride layers were then neutralized using 2 mL of saturated sodium bicarbonate. Next, the organic layer was isolated and washed with 2 mL of 5% sodium bisulfate and 2 mL of water. The organic solution was then dried with two spatial scoops of

anhydrous sodium sulfate. The DCM was then evaporated over a water bath using a watch glass and the crystals were weighed to calculate a percent yield.

Results

Molecular Weight of Borneol (g/mol)	Molecular Weight of Camphor (g/mol)
154.25	**152.23**

Table 1. Molecular Weight of Borneol and Camphor in g/mol

Theoretical Yield of Camphor (g)	Actual Yield of Camphor (g)
0.296	**0.217**

Table 2. Theoretical and Actual Yield of Camphor in g

Calculations

$$0.3g\ Borneol\ \div\ \frac{154.25g}{mol} = 0.001944895mol\ Borneol$$

$$0.001944895mol\ Borneol\ \times\ \frac{152.23g}{mol} =\ 0.296071313g\ Camphor$$

$$\frac{0.217g\ Camphor}{0.296g\ Camphor} \times 100 = 73.3\%$$

Discussion

In this experiment, borneol went through a series of reactions to produce a final product of camphor. Throughout the experiment two distinct layers will have formed, the bottom one being the organics layer containing the camphor product. The DCM (methylene chloride) used in the experiment has a density of 1.33g/mL which is greater than that of water thus the organics layer will be on the bottom. The additions of bleach made during the experiment are to be carried out dropwise because of the risk that an exothermic reaction may pose. Special attention is to be given while heating the mixture with the hot water bath to avoid superheating the product. During the neutralization step of the procedure, red litmus paper can be utilized to confirm the

reaction has been completed. Potential sources of error include the risk of superheating the mixture during the heating phase of the experiment and failure to let the DCM completely evaporate before weighing the final product. The primary source of error is likely the failure to let the DCM evaporate before taking the final weight of the camphor. To fix this issue, patiently waiting for all of the DCM to evaporate before measuring the final product would suffice. Not allowing the DCM to evaporate completely from the product will give a higher yield potentially giving a false percentage yield reading at the end of the experiment.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.